

Python Programming

Unit 02 – Lecture 03: Sets and Dictionaries

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Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

- Create and use sets for uniqueness and membership

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- Perform common set operations

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- Build dictionaries for key-value modeling

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- Create and use sets for uniqueness and membership
- Perform common set operations
- Build dictionaries for key-value modeling
- Sort and summarize collections with dicts

Sets: The Idea

- Unordered collection of unique elements

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- Fast membership testing

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- Unordered collection of unique elements
- Fast membership testing
- Useful for de-duplication

Set Creation and Membership

- Create with braces or `set()`

```
colors = {"red", "blue", "red"}  
print(colors)           # duplicates removed  
print("red" in colors)  # True
```

Set Creation and Membership

- Create with braces or `set()`
- Use `in` to test membership

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colors = {"red", "blue", "red"}  
print(colors)           # duplicates removed  
print("red" in colors)  # True
```

Set Operations

■ Union: $A \cup B$

Set Operations

- Union: $A \cup B$
- Intersection: $A \cap B$

Set Operations

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- Intersection: $A \cap B$
- Difference: $A - B$

Set Operations

- Union: $A \mid B$
- Intersection: $A \& B$
- Difference: $A - B$
- Symmetric difference: $A \hat{=} B$

Pitfall: Sets Are Unordered

- Sets do not support indexing (no `s[0]`)

```
s = {3, 1, 2}
print(sorted(s))      # [1, 2, 3]
```

Pitfall: Sets Are Unordered

- Sets do not support indexing (no `s[0]`)
- Use sets for membership/uniqueness, not ordering

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s = {3, 1, 2}
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```

Pitfall: Sets Are Unordered

- Sets do not support indexing (no `s[0]`)
- Use sets for membership/uniqueness, not ordering
- Convert to a sorted list when you need order

```
s = {3, 1, 2}
print(sorted(s))      # [1, 2, 3]
```

Dictionaries: Key-Value Model

- Map keys to values

```
student = {"name": "Asha", "marks": 88}  
print(student["name"])
```

Dictionaries: Key-Value Model

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- Keys must be immutable

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Dictionaries: Key-Value Model

- Map keys to values
- Keys must be immutable
- Fast lookup by key

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Dictionary Operations

- Access: `d[key]` or `d.get(key)`

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- Access: `d[key]` or `d.get(key)`
- Update: `d[key] = value`
- Iterate: `for k, v in d.items()`

Worked Example: Frequency Count Pattern

```
text = "to be or not to be"
freq = {}
for w in text.split():
    freq[w] = freq.get(w, 0) + 1
print(freq)
```

- `get(w, 0)` avoids `KeyError` for new words

Sorting Collections

- Sort dict items by key or value

```
counts = {"a": 3, "b": 1, "c": 2}
by_value = sorted(counts.items(), key=lambda x: x[1])
```

Sorting Collections

- Sort dict items by key or value
- `sorted(d.items(), key=...)`

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counts = {"a": 3, "b": 1, "c": 2}
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```

Applications

- Frequency counts with dictionaries

Applications

- Frequency counts with dictionaries
- Unique vocabulary with sets

Applications

- Frequency counts with dictionaries
- Unique vocabulary with sets
- Fast membership checks

Demo: Word Frequency and Unique Words

- Split a sentence into words

Script: `demo/word_frequency.py`

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- Split a sentence into words
- Build a frequency dictionary

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Demo: Word Frequency and Unique Words

- Split a sentence into words
- Build a frequency dictionary
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Demo: Word Frequency and Unique Words

- Split a sentence into words
- Build a frequency dictionary
- Build a set of unique words
- Extension: print top 3 most frequent words

Script: `demo/word_frequency.py`

Checkpoint 1

What is the result of `set([1,1,2,2])`?

Checkpoint 2

What is the difference between `d[key]` and `d.get(key)`?

Think-Pair-Share

Choose set vs list for membership checks and justify.

Summary

- Sets store unique elements and support set algebra

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- Dictionaries map keys to values for fast lookup/counting

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- Sets store unique elements and support set algebra
- Dictionaries map keys to values for fast lookup/counting
- Sorting dict items enables ranking and reporting

Exit Question

Write one line to sort dictionary items by value in descending order.